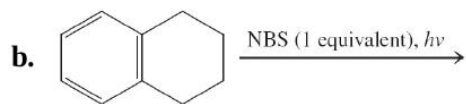
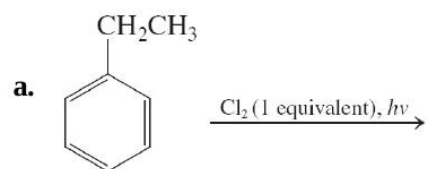


37. Give the expected major product(s) of each of the following reactions.



- 38.** Formulate a mechanism for the reaction described in [Problem 37](#), part (b).

39. Propose syntheses of each of the following compounds, beginning in each case with ethylbenzene. **(a)** (1-Chloroethyl)- benzene; **(b)** 2-phenylpropanoic acid; **(c)** 2-phenylethanol; **(d)** 2-phenyloxacyclopropane.

- 40.** Predict the order of relative stability of the three benzylic cations derived from chloromethylbenzene (benzyl chloride), 1-(chloromethyl)-4-methoxybenzene (4-methoxybenzyl chloride), and 1-(chloromethyl)-4-nitrobenzene (4-nitrobenzyl chloride). Rationalize your answer with the help of resonance structures.

41. By drawing appropriate resonance structures, illustrate why halogen atom attachment at the para position of the phenylmethyl (benzyl) radical is unfavored compared with attachment at the benzylic position.

CH 3

42. Triphenylmethyl radical, $(\text{C}_6\text{H}_5)_3\text{C}^\cdot$, is stable at room temperature in dilute solution in an inert solvent, and salts of triphenylmethyl cation, $(\text{C}_6\text{H}_5)_3\text{C}^+$, can be isolated as stable crystalline solids. Propose explanations for the unusual stabilities of these species.

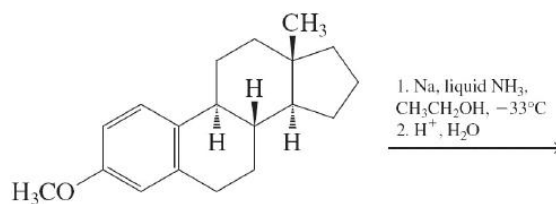
- 43.** Give the expected products of the following reactions or reaction sequences.

44. The hydrocarbon with the common name fluorene is acidic enough ($pK_a \approx 23$) to be a useful indicator in deprotonation reactions of compounds of greater acidity. Indicate the most acidic hydrogen(s) in fluorene. Draw resonance structures to explain the relative stability of its conjugate base.



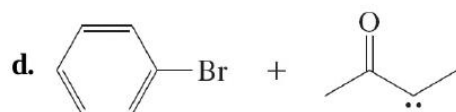
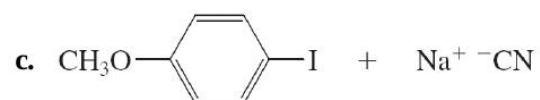
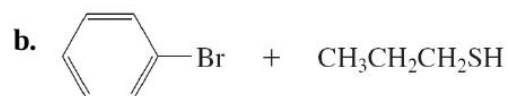
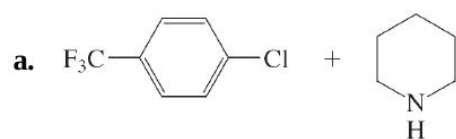
Fluorene

45. The Birch reduction shown below is widely used in the synthesis of a variety of highly biologically active steroids. Although the ^1H NMR spectrum of its product is complex, it exhibits a singlet at $\delta = 5.73$ ppm integrating to one hydrogen. In addition, the IR spectrum shows absorptions at $\tilde{\nu} = 1620$ and 1660 cm^{-1} , of medium and very strong intensities, respectively.



Propose a structure for the product and rationalize its formation

53. Give the likely products of each of the following reactions. Each one is carried out in the presence of a Pd catalyst, a phosphine, and heat.



54. A very efficient short synthesis of resveratrol ([Real Life 22-1](#)) was reported in 2006. Fill in reasonable reagents for steps (a) through (d). Refer to the referenced text sections as needed.

66. As mentioned in [Section 22-10](#), the Sandmeyer reactions, which entail the substitution of the nitrogen in arenediazonium salts by Cl, Br, CN , require copper(I) ion as a catalyst and proceed by complex radical mechanisms. Why do such substitutions not follow an $\text{S}_{\text{N}}2$ pathway? Why is the $\text{S}_{\text{N}}1$ mechanism, which operates in the corresponding displacements with OH and I, not effective?